

4-4 Investigation (for Loops)

1. **Count to Twenty:** Use a **for** loop to print the numbers from 1 to 20, inclusive.
2. **One Million:** Use a **for** loop to print the numbers from one to one million.
3. **Summing a Million:** Use a **for** loop to calculate the sum of all numbers from one to one million.
4. **Odd Numbers:** Make a range of the odd numbers from 1 to 20. Use a **for** loop to print each number.
5. **Threes:** Make a range of the multiples of 3 from 3 to 30. Use a **for** loop to print the numbers.
6. **Cubes:** A number raised to the third power is called a *cube*. For example, the cube of 2 is written as `2**3` in Python. Use a range and a **for** loop to print out the cube of each integer from 1 through 10.
7. **Pizzas:** Think of at least three kinds of your favorite pizza. Store these pizza names in a list, and then use a **for** loop to print the name of each pizza.
 - a. Modify your **for** loop to print a sentence using the name of each pizza instead of printing just the name of the pizza. For each pizza you should have one line of output containing a simple statement like *I like pepperoni pizza*.
 - b. Add a line at the end of your program, outside the loop, that states how much you like pizza. The output should consist of three or more lines about the kinds of pizza you like and then an additional sentence, such as *I really love pizza!*
8. **Animals:** Think of at least three different animals that have a common characteristic. Store the names of these animals in a list, and then use a **for** loop to print out the name of each animal.
 - a. Modify your program to print a statement about each animal, such as *A dog would make a great pet*.
 - b. Add a line at the end of your program stating what these animals have in common. You could print a sentence such as *Any of these animals would make a great pet!*